

SAI AKSHAY MENTA

(617) 406-7914 | akshaymenta24@gmail.com | linkedin.com/in/akshay-menta | github.com/akshay-menta

SUMMARY

AI/ML engineer with an Master's degree in AI from Northeastern, building Python data pipelines, document ingestion systems, and embedding based search on Azure. Strong in NLP, Model Finetuning, vector similarity and taking systems from proof of concept to production.

EDUCATION

Northeastern University

Master of Science in Artificial Intelligence [GPA:3.8/4.0]

Sep 2024 - Apr 2026

Boston, MA

Coursework: Natural Language Processing, Large Language Models, Machine Learning, Advanced Perception

Amrita Vishwa Vidyapeetham University

Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) [GPA:3.7/4.0]

Oct 2020 - Jun 2024

Coimbatore, India

Coursework: Linear Algebra, Probability & Statistics, Advanced Reinforcement Learning, Big Data Analytics

SKILLS

Languages	Python, C++, SQL, JavaScript
Data Engineering	ETL/Data Pipelines, Pandas, NumPy, PostgreSQL, ChromaDB, Power BI
ML/DL	Numpy, PyTorch, NLP, LLM Fine-tuning, Multimodal LLMs, Machine Learning, Computer Vision
Frameworks	Hugging Face, LangChain, CrewAI, Scikit-learn, FastAPI, Flask, Streamlit, React, OpenCV
Cloud & Tools	AWS, Azure, Docker, Kubernetes, Git, CI/CD, Ollama, Power Automate, Fabric

EXPERIENCE

Machine Learning Engineer/Researcher

Apr 2025 - Present

InsightX Lab - Northeastern University

Boston, MA

- Built an end-to-end Python data pipeline that processes 2.1GB of Elliptic++ Bitcoin blockchain data into PyTorch Geometric graph objects, handling 1M+ nodes and 3.1M+ edges with temporal splits and class-weighted sampling to address 2% illicit class imbalance, completing in 37 seconds.
- Architected a multi-agent fraud detection system that combines GNN-based transaction and wallet scoring with rule-based compliance checks to produce auditable, human-readable compliance reports.

Machine Learning Engineer

Feb 2024 - Jul 2024

Genpact

Hyderabad, India

- Built an embedding-based information retrieval system in Python to automate document search and response generation, improving response accuracy by 25% and cutting manual lookup time by 40%.
- Developed reliable Python data pipelines with validation, logging, and error handling across SQL data sources to support finance and operations workflows.
- Built interactive Power BI dashboards with SQL-driven data models for sales and pricing analysis, reducing reporting turnaround by 30% and enabling data-driven decisions across the team.

PROJECTS

InfoMeme: AI-Driven Meme Generation from News

Sep 2025 - Dec 2025

Tech Stack: Python, AWS (S3, EC2 GPU), OpenAI GPT-4 Vision, FAISS, Streamlit

- Built an end-to-end multimodal data pipeline in Python combining LLM-based text generation with FAISS-powered semantic retrieval across 10,000+ news embeddings, cutting content matching latency by 35%.
- Automated the full workflow from news ingestion to meme output on AWS, using S3 for embedding and asset storage and EC2 GPU instances for inference, demonstrating hands-on experience with cloud-based agentic pipelines and scalable inference systems.

EmoLingo: Multi-Model Document Classification System

Jan 2025 - May 2025

Tech Stack: Python, Hugging Face Transformers, RoBERTa, T5, ELECTRA, DistilBERT

- Built a Python pipeline that fine tuned 3 transformer models to classify documents across 43 labels (emotion, tone, proficiency), achieving 97% Hamming accuracy, 77.78% tone accuracy, and 0.66 macro F1 on proficiency.
- Trained on 140,000+ samples with augmentation to rebalance minority classes, and shipped the system as a modular Streamlit service where each classifier can be retrained or swapped independently.

PromptPrint: Multi-Agent LLM Optimization System

MIT AI Hackathon – Jun 2025

Tech Stack: Python, Langchain, OpenRouter, LLM APIs, Energy Estimation APIs

- Built an agentic framework in Python that routes user queries across 15+ LLMs using reinforcement-based agent coordination, balancing latency, cost, and real-time carbon footprint scoring per request.
- Achieved 40% reduction in energy consumption versus a single-model baseline while maintaining response quality, demonstrating viable sustainability-aware LLM serving at the agentic layer.

CERTIFICATIONS & PUBLICATIONS

- Microsoft Certified: Azure AI Fundamentals [\[Link\]](#)
- Monday AI pathway Level II: AI agents [\[Link\]](#)
- Improving Reinforcement Learning Agent Training using Text based Guidance, ACL Anthology [\[Link\]](#)
- A Few-Shot Approach to Dysarthric Speech Intelligibility Level Classification Using Transformers, 14th ICCNT [\[Link\]](#)
- Enhancing Knee Osteoarthritis Severity Level Classification Using Diffusion Augmented Images, Springer ICACECS [\[Link\]](#)
- Semi Supervised Flood Damage Detection Using Satellite Images, Springer Nature, ICCAIML [\[Link\]](#)